

Recognise the effect of using approximations of real numbers in repeated calculations and compare the results when using exact representations

Learning intention

- recognise the effect of using approximations of real numbers in repeated calculations and compare the results when using exact representations.

Teach and model

- investigating the impact of approximation on multiple calculations in contexts that involve the area of compound shapes involving circles.
- comparing and contrasting the effect of truncation or rounding on the final result of calculations when using approximations of real numbers rather than exact representations.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.